

Dr. M. Nur Hasan

Computational Materials Physicist

Electronic Structure • Magnetism • Multiscale Materials Modelling

Materials Theory, Department of Physics and Astronomy

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Professional Profile

Computational materials physicist specializing in first-principles electronic-structure theory and computational magnetism. Expertise in DFT, magnetic anisotropy, exchange interactions, finite-temperature magnetism, and atomistic spin dynamics, with applications to permanent magnets, quantum materials, and low-dimensional magnetic systems. Currently a Postdoctoral Researcher in the Department of Physics and Astronomy at Uppsala University within the EU Horizon Europe MaMMoS project.

Research Areas

Computational magnetism • First-principles electronic structure theory • Magnetic anisotropy and exchange interactions • Finite-temperature magnetism • Atomistic spin dynamics • Permanent magnets • Correlated quantum materials • 2D and van der Waals magnetism • Spintronics and chiral magnetism

Professional Experience

Postdoctoral Researcher

[April 2024 – Present]

Materials Theory, Department of Physics and Astronomy, Uppsala University, Sweden
EU Horizon Europe MaMMoS Project

First-principles and multiscale modelling of permanent magnets, linking electronic structure, magnetic anisotropy and exchange interactions to finite-temperature magnetism, atomistic spin dynamics, and micromagnetic performance within the EU Horizon Europe MaMMoS project.

Doctoral Researcher (CSIR fellow)

[2017–2023]

S. N. Bose National Centre for Basic Sciences, Kolkata, India

First-principles studies of correlated and magnetic quantum materials, including kagome metals, van der Waals magnets, and spintronic systems, leading to publications in *Physical Review Letters*, *Physical Review B* and *npj 2D Materials and Applications*.

Selected Research Highlights

Permanent Magnets: First-principles and multiscale modelling of magnetic materials, combining chemical substitution and disorder modelling with magnetic anisotropy, exchange interactions, finite-temperature magnetism, atomistic spin dynamics, and micromagnetics for computational design and optimization of high-performance permanent magnets. *IEEE Trans. Magn.* (2026); *arXiv:2609.11464* (2026).

Two-Dimensional and van der Waals Magnetism: First-principles study of low-dimensional magnets, particularly Fe_4GeTe_2 and 1T-CrS_2 . *arXiv:2607.23625* (2026); *ACS Nano* **20**, 24621 (2026); *Phys. Rev. B* **111**, 134449 (2025).

Correlated Quantum Materials: Study of correlation-driven magnetism, magnetic fluctuations, and exchange interactions in kagome AV_3Sb_5 ($A = \text{K, Rb, Cs}$). *Phys. Rev. Lett.* **131**, 196702 (2023); *Phys. Rev. B* **108**, 174413 (2023).

Spintronics and Chiral Magnetism: First-principles study of interfacial Dzyaloshinskii–Moriya interaction, spin–orbit effects, and chiral magnetism in magnetic heterostructures. *ACS Nano* **19**, 23564–23574 (2025).

Technical Expertise

Electronic Structure: Density functional theory (DFT), correlated electronic-structure methods, band structures, density of states, Fermi-surface analysis, and high-throughput first-principles calculations.

Computational Magnetism: Magnetocrystalline anisotropy, exchange interactions, Dzyaloshinskii–Moriya interaction (DMI), anisotropic exchange, finite-temperature magnetism, Monte Carlo simulations, atomistic spin dynamics, spin textures, and chiral magnetism.

Materials Modelling: Chemical disorder and special quasirandom structures (SQS), structural optimization, phonons and vibrational free energies, phase stability, finite-temperature modelling, machine-learning interatomic potentials (MLIPs), and magnetic-materials design.

Software: VASP, RSPt, UppASD, ATAT, Phonopy, MatterSim, Wannier90, WannierTools, WannierBerri, ASE, IFermi, PyProcar, FermiSurfer, VESTA, and XCrySDen.

Programming & HPC: Python, Bash/Shell scripting, workflow automation, SLURM, MPI, and large-scale parallel computing on national and European HPC systems, including Dardel, Tetralith, LUMI, Pelle/UPPMAX, and Arrhenius.

Selected Publications

Sk Md Obaidulla, **M. N. Hasan**, D. Das, R. Alam, A. Supina, M. A. Aslam, S. Kamal, I. Šarić Jankovic, A. Matkovic, C. Teichert, A. N. Pal, H. C. Herper, & M. Kralj; *Observation of Correlation-Driven Topological Transport and Robust Ferromagnetism in 2D CrS₂*. *arXiv:2607.23625* (2026).

R. Pal, **M. N. Hasan**, D. Das, C. Nayak, M. Deka, N. Salehi, M. Pereiro, S. Mondal, A. Misra, A. Singha, P. Mandal, D. Karmakar, A. N. Pal; *Spin Reorientation Driven Renormalization of Spin–Phonon Coupling in Fe₄GeTe₂*. *ACS Nano* **20**, 24621 (2026).

M. N. Hasan, N. Salehi, F. Sorgenfrei, A. Delin, I. Di Marco, A. Bergman, M. Pereiro, P. Thunström, O. Eriksson, & D. Karmakar; Dynamical electronic correlations and chiral magnetism in the van der Waals magnet Fe₄GeTe₂. *Phys. Rev. B* **111**, 134449 (2025).

S. Pal, **M. N. Hasan**, H. Bangar, N. Salehi, S. Mathur, M. Pereiro, P. Thunström, P. K. Muduli, D. Karmakar, & A. Barman; Deciphering Interfacial Dzyaloshinskii–Moriya Interaction in Transition-Metal Dichalcogenide/Permalloy Heterostructures by Brillouin Light Scattering and First-Principles Calculations. *ACS Nano* **19**, 23564–23574 (2025).

M. N. Hasan, R. Bharati, J. Hellsvik, A. Delin, S. K. Pal, A. Bergman, S. Sharma, I. Di Marco, M. Pereiro, P. Thunström, P. M. Oppeneer, O. Eriksson, & D. Karmakar; Magnetism in AV₃Sb₅ (Cs, Rb and K): Origin and Consequences for the Strongly Correlated Phases. *Phys. Rev. Lett.* **131**, 196702 (2023).

33+ publications • 620+ citations • h-index: 12 • [Check full list of publications](#)

Scientific Activities

Co-organizer, European School on Magnetism (ESM 2026) [2026]
Contributed to the organization of ESM 2026 at Uppsala University, Uppsala, Sweden.

Co-organizer & Oral Presenter, DQMMA 2023 [2023]
First Indo–Swedish Meeting on *Divergent Quantum Materials, Methods and Applications*, Goa, India. Contributed to the scientific organization and delivered an oral presentation.

Conference Presentations: DPG Spring Meeting (Oral)–2025; JEMS (Poster)–2025; IEEE-AIM (Oral)–2025; APS March Meeting (Oral)–2023; Psi-k Conference (Oral & Poster)–2022; among others.

Education

Doctor of Philosophy (Ph.D.), Physics [December 2023]
S. N. Bose National Centre for Basic Sciences, Kolkata, India
Awarded by the University of Calcutta, India.

Master of Science (M.Sc.), Physics [2015 - 2017]
Department of Physics, Aligarh Muslim University, Aligarh - 202002, India.

Bachelor of Science in Physics [2012 - 2015]
Department of Physics, Aligarh Muslim University, Aligarh - 202002, India.

Computational Resources, Awards & Fellowships

NAISS / Arrhenius Computational Resource Award, Sweden [2026]
Awarded HPC resources through an independently submitted research proposal for large-scale electronic-structure and computational-magnetism simulations.

UPPMAX Computational Resource Award, Sweden [2026]
Awarded HPC resources through an independently submitted research proposal for first-principles and computational-magnetism simulations.

SERB International Travel Grant [2022]
CSIR Foreign Travel Grant [2022]
CSIR Senior Research Fellowship (SRF) [2020–2023]
CSIR Junior Research Fellowship (JRF) [2018–2020]
CSIR-NET Junior Research Fellowship [2017]
